

# **SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE**

**SUBJECT NAME: T101 Cyber & Information Security Practical**

## **PRACTICAL - 1**

### **Practical 1(A-1)**

**Aim :-** To design and implement an algorithm to encrypt and decrypt messages using the Classical Substitution Technique (Caesar Cipher).

#### **Code :-**

#To design and implement an algorithm to encrypt and decrypt messages using the Classical Substitution Technique (Caesar Cipher).

```
def encrypt(text, shift):
```

```
    result = ""
```

```
    for ch in text:
```

```
        if ch.isalpha():
```

```
            if ch.isupper():
```

```
                result += chr((ord(ch) - 65 + shift) % 26 + 65)
```

```
            else:
```

```
                result += chr((ord(ch) - 97 + shift) % 26 + 97)
```

```
        else:
```

```
            result += ch
```

```
    return result
```

```
def decrypt(text, shift):
```

```
    return encrypt(text, -shift)
```

```
message = input("Enter Message : ")
```

```
shift = int(input("Enter Shift Value : "))
```

```
encrypted = encrypt(message, shift)
```

```
print("Encrypted Message :", encrypted)
```

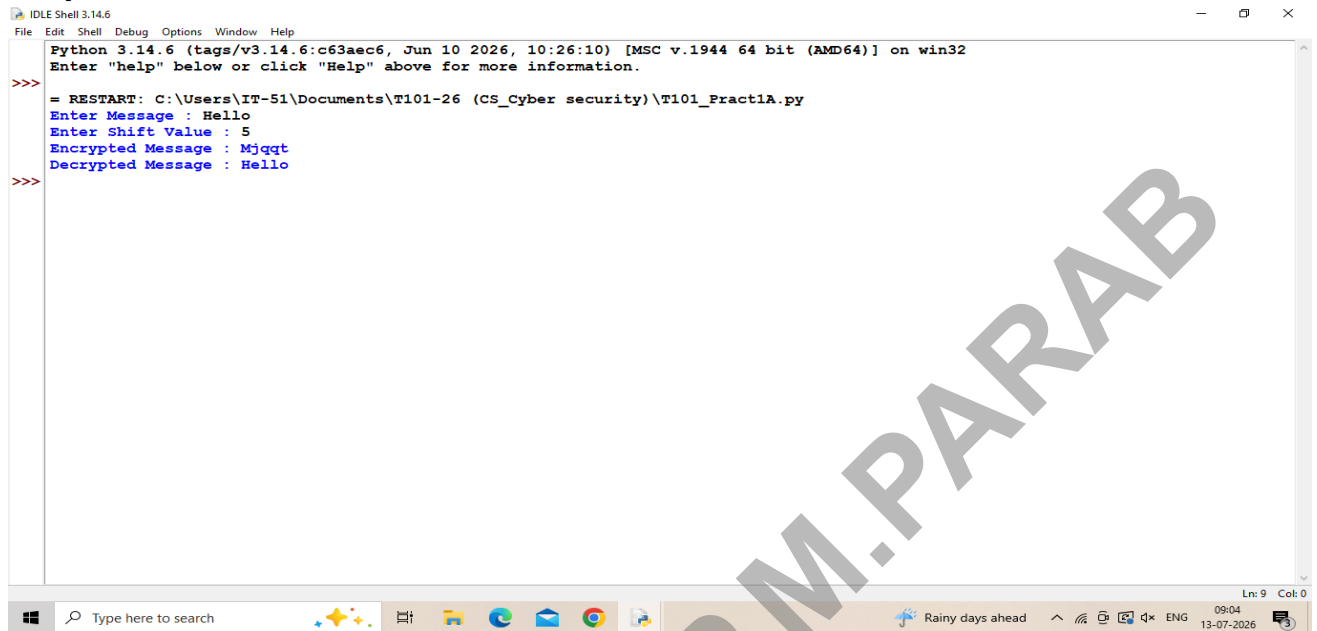
```
decrypted = decrypt(encrypted, shift)
```

```
print("Decrypted Message :", decrypted)
```

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Output :-



```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:\Users\IT-51\Documents\T101-26 (CS_Cyber security)\T101_Pract1A.py
Enter Message : Hello
Enter Shift Value : 5
Encrypted Message : Mjqqt
Decrypted Message : Hello
>>>
```

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Practical 1(A-2)

**Aim :-** To design and implement an algorithm to encrypt and decrypt messages using the Classical Substitution Technique (Caesar Cipher [GUI]).

## **Code :-**

```
# Design and implement algorithms to encrypt and decrypt messages using classical
substitution techniques.(GUI)
import tkinter as tk
from tkinter import messagebox
def encrypt_message():
    message = message_box.get()
    key_text = key_box.get()
    if message == "" or key_text == "":
        messagebox.showwarning("Input Error", "Please enter the message and key.")
        return
    try:
        key = int(key_text)
    except ValueError:
        messagebox.showerror("Error", "Key must be a number.")
        return
    encrypted = ""
    for ch in message:
        if ch.isalpha():
            if ch.isupper():
                encrypted += chr((ord(ch) - ord('A') + key) % 26 + ord('A'))
            else:
                encrypted += chr((ord(ch) - ord('a') + key) % 26 + ord('a'))
        else:
            encrypted += ch
    result.config(text=encrypted)
def decrypt_message():
    message = message_box.get()
    key_text = key_box.get()
    if message == "" or key_text == "":
        messagebox.showwarning("Input Error", "Please enter the message and key.")
        return
    try:
        key = int(key_text)
    except ValueError:
```

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```
messagebox.showerror("Error", "Key must be a number.")
return
decrypted = ""
for ch in message:
    if ch.isalpha():
        if ch.isupper():
            decrypted += chr((ord(ch) - ord('A') - key) % 26 + ord('A'))
        else:
            decrypted += chr((ord(ch) - ord('a') - key) % 26 + ord('a'))
        else:
            decrypted += ch
result.config(text=decrypted)
def clear_data():
    message_box.delete(0, tk.END)
    key_box.delete(0, tk.END)
    result.config(text="")
window = tk.Tk()
window.title("Caesar Cipher")
window.geometry("450x320")
window.resizable(False, False)
window.configure(bg="#f4f4f4")
title = tk.Label(
    window,
    text="Classical Substitution Cipher",
    font=("Arial", 16, "bold"),
    bg="#f4f4f4"
)
title.pack(pady=15)
tk.Label(
    window,
    text="Enter Message",
    font=("Arial", 11),
    bg="#f4f4f4"
).pack()

message_box = tk.Entry(window, font=("Arial", 12), width=35)
message_box.pack(pady=5)
tk.Label(
    window,
    text="Enter Key",
    font=("Arial", 11),
    bg="#f4f4f4"
```

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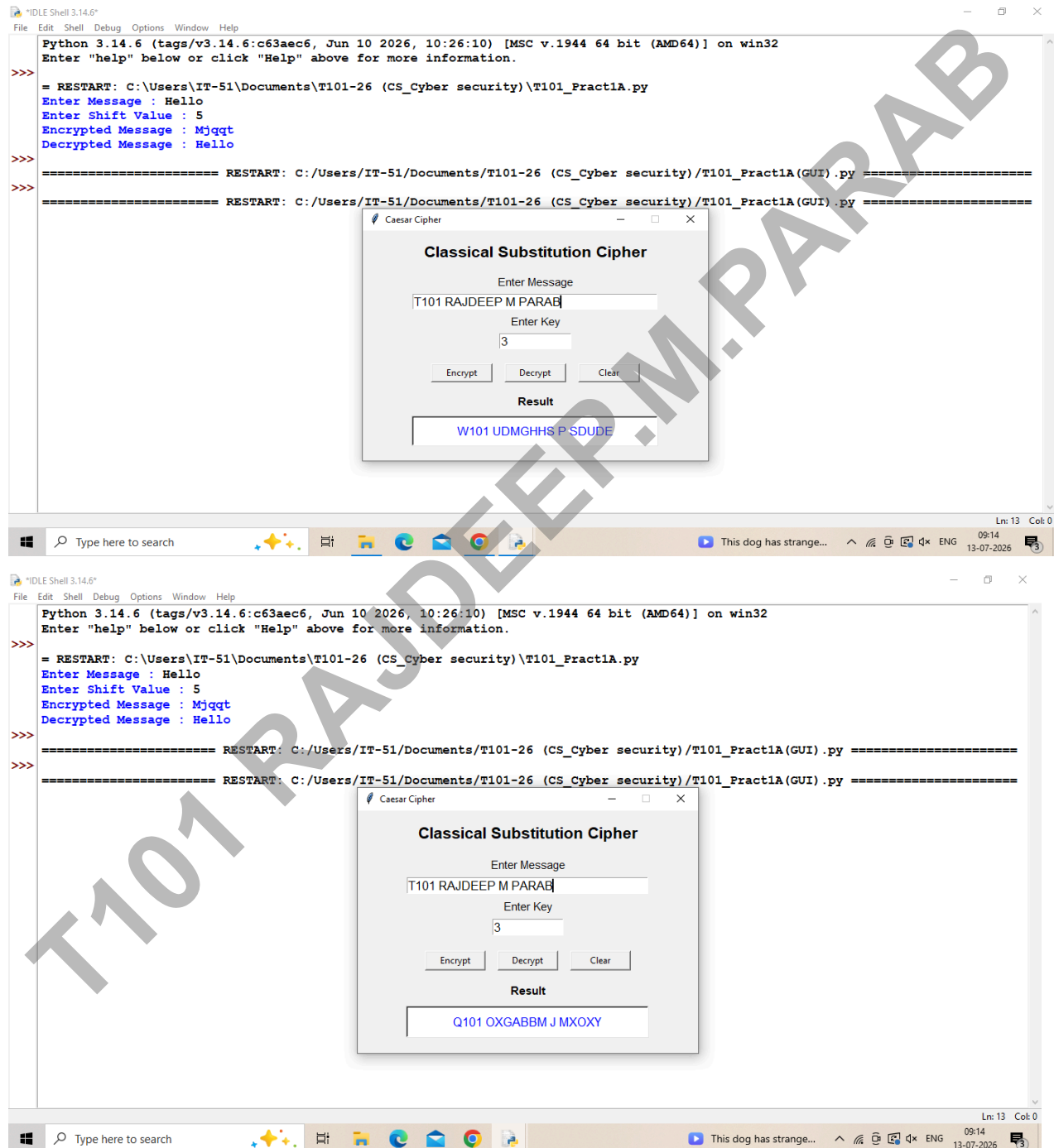
```
).pack()
key_box = tk.Entry(window, font=("Arial", 12), width=10)
key_box.pack(pady=5)
button_frame = tk.Frame(window, bg="#f4f4f4")
button_frame.pack(pady=15)
encrypt_btn = tk.Button(
    button_frame,
    text="Encrypt",
    width=10,
    command=encrypt_message
)
encrypt_btn.grid(row=0, column=0, padx=8)
decrypt_btn = tk.Button(
    button_frame,
    text="Decrypt",
    width=10,
    command=decrypt_message
)
decrypt_btn.grid(row=0, column=1, padx=8)
clear_btn = tk.Button(
    button_frame,
    text="Clear",
    width=10,
    command=clear_data
)
clear_btn.grid(row=0, column=2, padx=8)
tk.Label(
    window,
    text="Result",
    font=("Arial", 11, "bold"),
    bg="#f4f4f4"
).pack()
result = tk.Label(
    window,
    text="",
    font=("Arial", 12),
    fg="blue",
    bg="white",
    width=35,
    height=2,
    relief="sunken"
)
```

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```
result.pack(pady=8)
window.mainloop()
```

### Output :-



```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\IT-51\Documents\T101-26 (CS_Cyber security)\T101_Pract1A.py
Enter Message : Hello
Enter Shift Value : 5
Encrypted Message : Mjqqt
Decrypted Message : Hello
>>>
===== RESTART: C:/Users/IT-51/Documents/T101-26 (CS_Cyber security)/T101_Pract1A(GUI).py =====
>>>
===== RESTART: C:/Users/IT-51/Documents/T101-26 (CS_Cyber security)/T101_Pract1A(GUI).py =====
```

Caesar Cipher

Classical Substitution Cipher

Enter Message  
T101 RAJDEEP M PARAB

Enter Key  
3

Encrypt Decrypt Clear

Result  
W101 UDMGHHS P SQUDE

Caesar Cipher

Classical Substitution Cipher

Enter Message  
T101 RAJDEEP M PARAB

Enter Key  
3

Encrypt Decrypt Clear

Result  
Q101 OXGABBM J MXOXY

T101  
RAJDEEPP M PARAB  
TY.BSc.CS

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**Practical 1(B-1)**

**Aim :-** Design and implement algorithms to encrypt and decrypt messages using transposition techniques.

**Code :-**

#Design and implement algorithms to encrypt and decrypt messages using transposition techniques.

```
def encrypt(text):
    first = ""
    second = ""
    for i in range(len(text)):
        if i % 2 == 0:
            first += text[i]
        else:
            second += text[i]
    return first + second

def decrypt(text):
    mid = (len(text) + 1) // 2
    first = text[:mid]
    second = text[mid:]
    result = ""
    for i in range(mid):
        result += first[i]
        if i < len(second):
            result += second[i]
    return result

message = input("Enter message: ")
encrypted = encrypt(message)
print("\nEncrypted Message :", encrypted)
decrypted = decrypt(encrypted)
print("Decrypted Message :", decrypted)
```

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Output :-

```
IDLE Shell 3.14.6
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:\Users\IT-51\Documents\T101-26 (CS_Cyber security)\T101_Pract1A.py
Enter Message : Hello
Enter Shift Value : 5
Encrypted Message : Mjqqt
Decrypted Message : Hello

>>>
===== RESTART: C:/Users/IT-51/Documents/T101-26 (CS_Cyber security)/T101_Pract1A(GUI).py =====
>>>
===== RESTART: C:/Users/IT-51/Documents/T101-26 (CS_Cyber security)/T101_Pract1A(GUI).py =====
>>>
===== RESTART: C:/Users/IT-51/Documents/T101-26 (CS_Cyber security)/T101_Pract1B.py =====
Enter message: T101 RAJDEEP M PARAB
Encrypted Message : T0 ADE AA11RJEPMPRB
Decrypted Message : T101 RAJDEEP M PARAB

>>>
```



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**SUBJECT NAME: T101 Cyber & Information Security Practical**

**Practical 1(B-2)**

**Aim :-** Design and implement algorithms to encrypt and decrypt messages using transposition techniques (GUI).

**Code :-**

#Design and implement algorithms to encrypt and decrypt messages using the Transposition (Rail Fence) Cipher in GUI.

import tkinter as tk

def encrypt\_text():

text = message\_entry.get()

if text == "":

result\_label.config(text="Please enter a message.")

return

first\_row = ""

second\_row = ""

for i in range(len(text)):

if i % 2 == 0:

first\_row += text[i]

else:

second\_row += text[i]

encrypted = first\_row + second\_row

result\_label.config(text="Encrypted Message : " + encrypted)

def decrypt\_text():

text = message\_entry.get()

if text == "":

result\_label.config(text="Please enter a message.")

return

middle = (len(text) + 1) // 2

first\_row = text[:middle]

second\_row = text[middle:]

original = ""

for i in range(middle):

original += first\_row[i]

if i < len(second\_row):

original += second\_row[i]

result\_label.config(text="Decrypted Message : " + original)

def clear\_all():

message\_entry.delete(0, tk.END)

result\_label.config(text="")

window = tk.Tk()

**T101**

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```
window.title("Transposition Cipher")
window.geometry("500x300")
window.resizable(False, False)
heading = tk.Label(
    window,
    text="Transposition Cipher (Rail Fence - 2 Rails)",
    font=("Arial", 14, "bold")
)
heading.pack(pady=10)
frame = tk.Frame(window)
frame.pack(pady=10)
message_label = tk.Label(frame, text="Enter Message :", font=("Arial", 11))
message_label.grid(row=0, column=0, padx=10, pady=10)
message_entry = tk.Entry(frame, width=35, font=("Arial", 11))
message_entry.grid(row=0, column=1)
button_frame = tk.Frame(window)
button_frame.pack(pady=10)
encrypt_button = tk.Button(
    button_frame,
    text="Encrypt",
    width=12,
    command=encrypt_text
)
encrypt_button.grid(row=0, column=0, padx=10)

decrypt_button = tk.Button(
    button_frame,
    text="Decrypt",
    width=12,
    command=decrypt_text
)
decrypt_button.grid(row=0, column=1, padx=10)
clear_button = tk.Button(
    button_frame,
    text="Clear",
    width=12,
    command=clear_all
)
clear_button.grid(row=0, column=2, padx=10)
result_title = tk.Label(
    window,
    text="Result",
```

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**SUBJECT NAME: T101 Cyber & Information Security Practical**

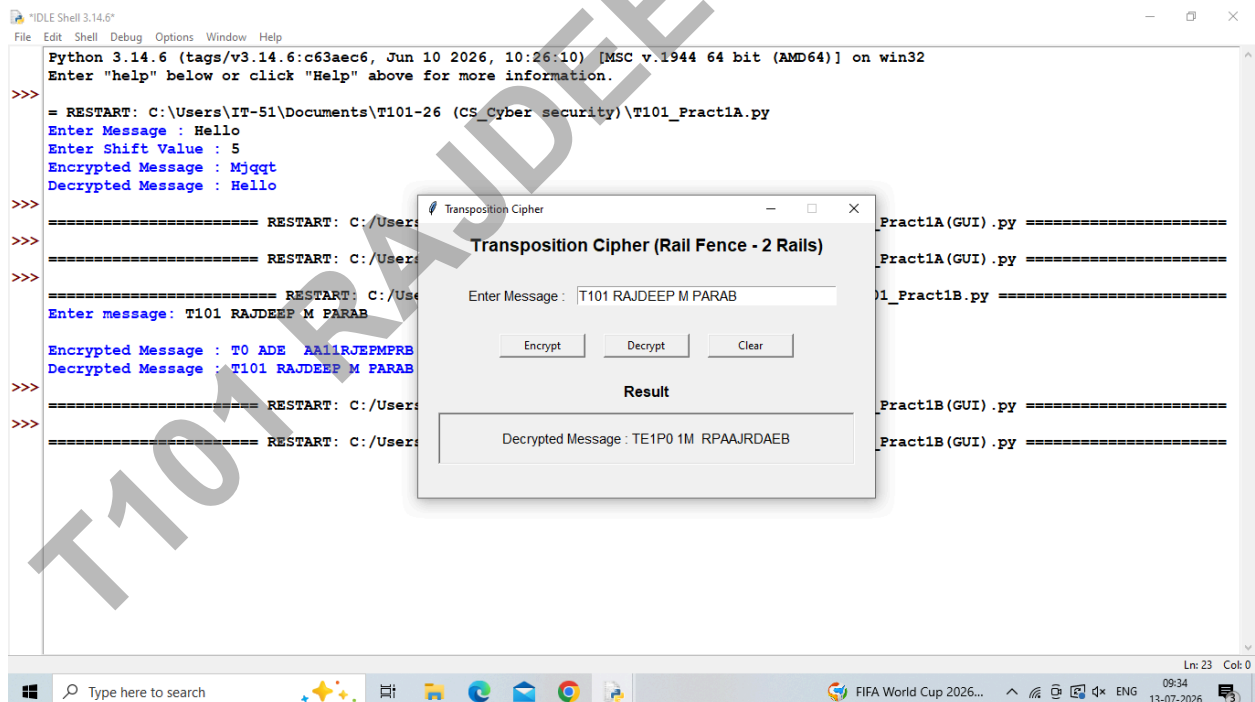
```
font=("Arial", 12, "bold")
)
result_title.pack(pady=(15, 5))
result_label = tk.Label(
    window,
    text="",
    width=50,
    height=3,
    relief="sunken",
    anchor="center",
    font=("Arial", 11)
)
result_label.pack(pady=5)
window.mainloop()
```

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The screenshot displays a Windows desktop environment. In the background, the IDLE Shell window is open, showing a Python script for a transposition cipher. The script includes a help message, a restart command, and a function to encrypt and decrypt messages using a rail fence cipher with 2 rails. The output shows the message 'T101 RAJDEEP M PARAB' being encrypted to 'T0 ADE AA11RJEPMPRB' and then decrypted back to 'T101 RAJDEEP M PARAB'.

In the foreground, a window titled 'Transposition Cipher' is open. It features a text input field containing 'T101 RAJDEEP M PARAB', three buttons labeled 'Encrypt', 'Decrypt', and 'Clear', and a 'Result' section. The 'Result' section displays the encrypted message: 'Encrypted Message : T0 ADE AA11RJEPMPRB'.

A large, diagonal watermark reading 'P.M.P.R.A.B' is visible across the center of the image.



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